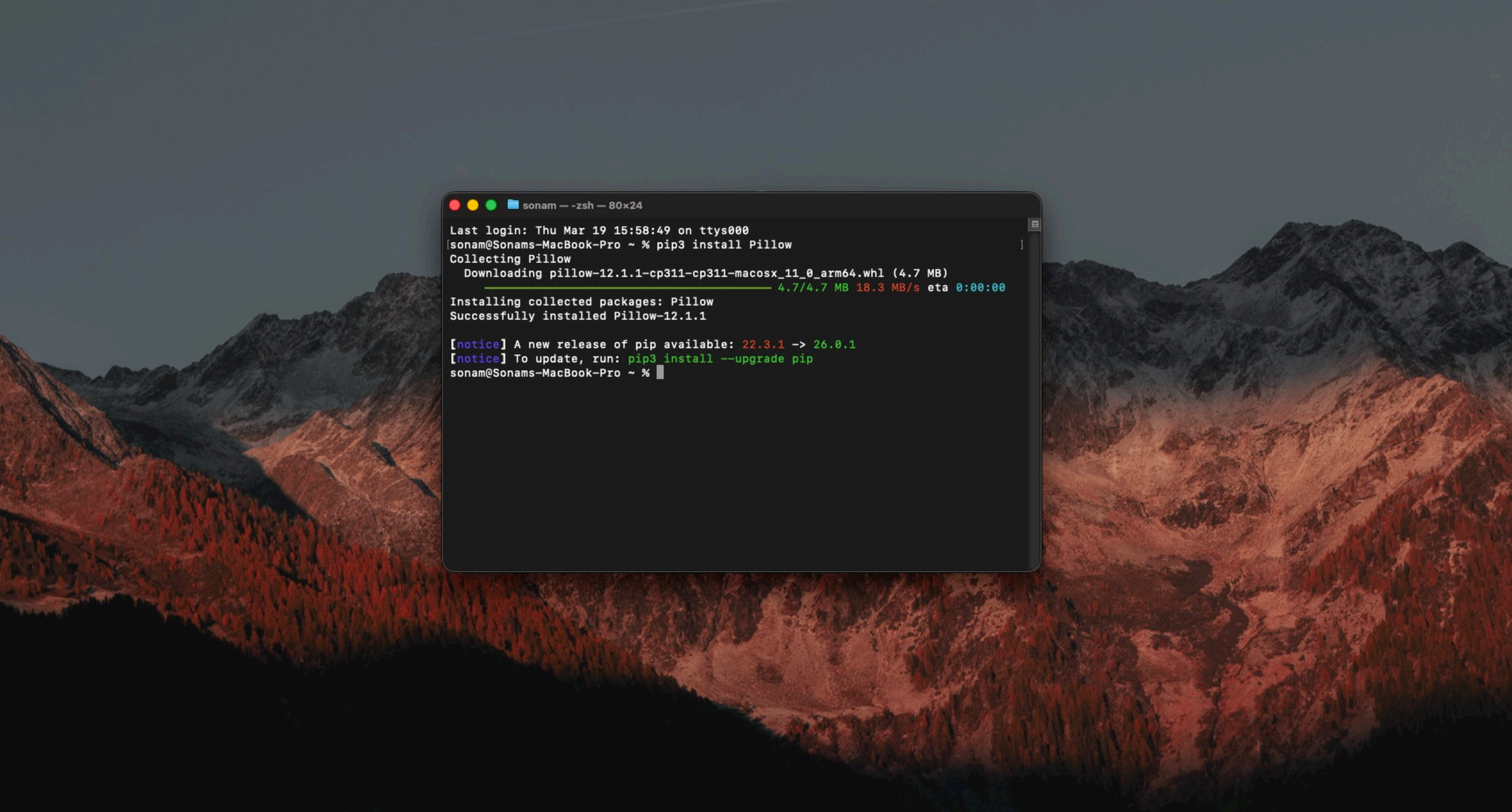

sonam — -zsh — 80×24
Last login: Thu Mar 19 15:58:49 on ttys000
sonam@Sonams-MacBook-Pro ~ % pip3 install Pillow
Collecting Pillow
Downloading pillow-12.1.1-cp311-cp311-macosx_11_0_arm64.whl (4.7 MB)
4.7/4.7 MB 18.3 MB/s eta 0:00:00
Installing collected packages: Pillow
Successfully installed Pillow-12.1.1
[notice] A new release of pip available: 22.3.1 -> 26.0.1
[notice] To update, run: pip3 install --upgrade pip
sonam@Sonams-MacBook-Pro ~ %

python_basics Version Control Current File Trial

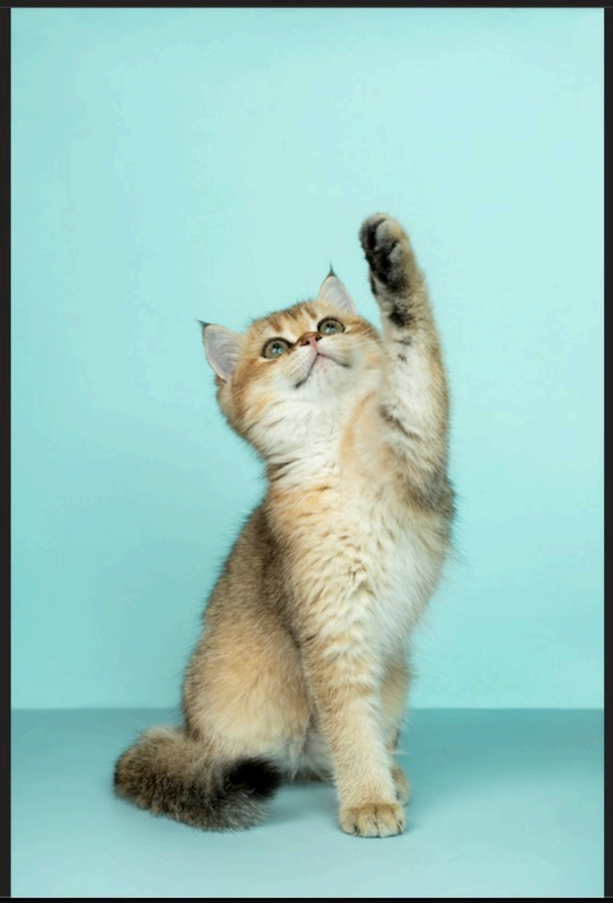
Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Insects
  - Week_4_Lab_Activities
  - 1_Gammar 2021_1_2021_
  - 1_Limniu 2021_1_2021_0
  - cat.jpg
  - dog.jpg
  - eraser.png
  - intro.py
  - main.py
  - pencil.png
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py SoftwareTechnology1 intro.py

```python
#Importing the Pillow Library
from PIL import Image

#To import an image, we use the "open" method of the Image module, which requires the path to the image file as an argument.
image = Image.open('cat.jpg')


#to see the image, we need to store it in a variable and then display it using the "show" method sof the Image object
image.show()
```

tmp5yczxqqs....



Run intro

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/intro.py

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Insects
  - Week_4_Lab_Activities
  - 1_Gammar 2021_1_2021_
  - 1_Limniu 2021_1_2021_0
  - cat.jpg
  - dog.jpg
  - eraser.png
  - intro.py
  - main.py
  - pencil.png
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py SoftwareTechnology1 intro.py

```python
#Importing the Pillow Library
from PIL import Image

#To import an image, we use the "open" method of the Image module, which requires the path to the image file as an argument.
image = Image.open('cat.jpg')


#to see the image, we need to store it in a variable and then display it using the "show" method sof the Image object
image.show()

#this code will print the size of the image as a tuple (width, height), the name of the image file, and the file format.
print(image.size)
print(image.filename)
print(image.format)
```

Run intro

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/intro.py
(1920, 2879)
cat.jpg
JPEG

Process finished with exit code 0
```

tmpnpdxhbxc....

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - img_rotated.png
  - intro.py
  - main.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py SoftwareTechnology1 intro.py

```python
#intro.py
from PIL import Image

# create an image via import
image = Image.open('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG')

# analyze the image
print(image.size)
print(image.filename)
print(image.format)

#to flip the image
image = image.transpose(Image.Transpose.ROTATE_90)

#to show the image
image.show()

# exercise
#this will rotate the image 30 degree and save the output image.
img_rotated = image.rotate(30)
img_rotated.save( fp: 'img_rotated.png',  format: 'png')
```

tmppzx9xv5c....

PythonProject

Recent
Shared

Favourites
Applications
Downloads
Movies
Pictures
Music
Creative Cloud File...
Desktop
Documents

main.py intro.py SoftwareTechnology1 Week_4_Lab_Activities Assets img_rotated.png

Macintosh HD › Users › sonam › PycharmProjects › PythonProject

Run intro

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/intro.py
(464, 464)
Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG
PNG

Process finished with exit code 0
```

PythonProject › intro.py 18:11 LF UTF-8 4 spaces Python 3.11 (PythonProject)

python_basics Version Control Current File Trial

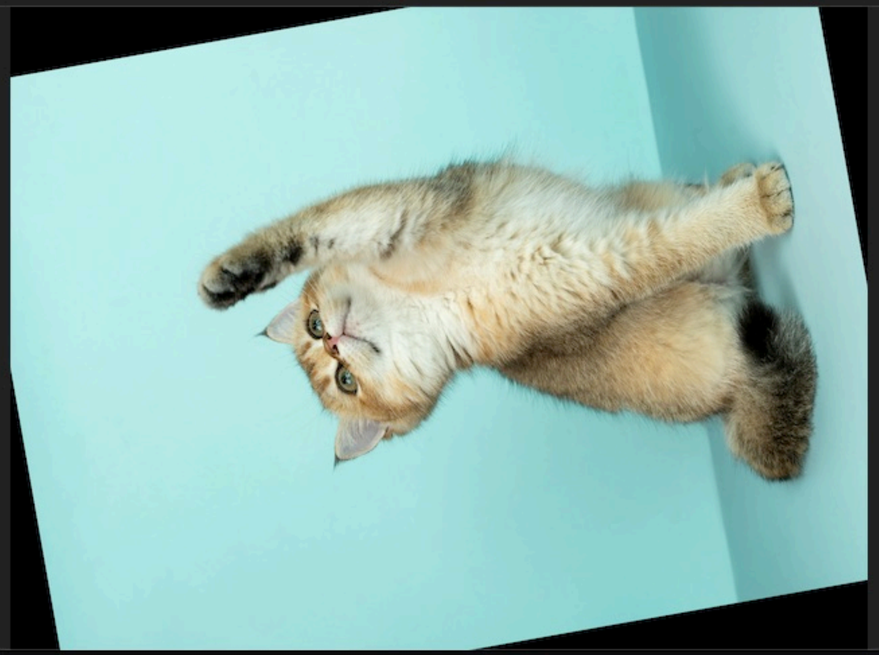
Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - img_rotated.png
  - intro.py
  - main.py
  - manipulate.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py SoftwareTechnology1 intro.py manipulate.py

```python
#manipulate.py
from PIL import Image

# import an image
image = Image.open('Assets/cat.jpg')

# flip
image_transpose = image.transpose(Image.Transpose.ROTATE_90)

# rotation
image_rotate = image.rotate( angle: 45, expand = False, center = (0,0))

# crop
image_crop = image.crop((800,600,1600,1600))

# resize
image_resize = image.resize((1000,600))

# applying multiple transformations to an image in a single line of code
combined_image = image.transpose(Image.Transpose.ROTATE_90).resize((2000,1500)).rotate(10)
combined_image.show()
```

tmpk0ip2m1i....



Run manipulate

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/manipulate.py

Process finished with exit code 0
```

PythonProject > manipulate.py 6:1 LF UTF-8 4 spaces Python 3.11 (PythonProject)

python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - explore_colors.py
  - img_rotated.png
  - intro.py
  - main.py
  - manipulate.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py | SoftwareTechnology1 | intro.py | manipulate.py | explore_colors.py

```python
# explore_colors.py
from PIL import Image

image = Image.open('Assets/eraser.png')

# get one pixel
print(image.getpixel((0,0)))

# # grayscale images
red_grayscale_image = image.getchannel('R')
blue_grayscale_image = image.getchannel('B')

red_grayscale_image.show()
blue_grayscale_image.show()

# change the pixel color
image.putpixel( xy: (0,0), value: (255,0,255))
image.show()

# explore
print(image.getpixel((0,0)))
for x in range(image.size[0]):
    for y in range(image.size[1]):
        if image.getpixel((x,y))[0] == 0:
            image.putpixel( xy: (x,y), value: (200,20,20))
image.show()
```

tmpnnzxw217....

tmp827_e328....

tmpw5uizqtv....

tmplr_zma6d....

Run explore_colors

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/explore_colors.py
(255, 255, 255, 255)
(255, 0, 255, 255)

Process finished with exit code 0
```

PythonProject > explore_colors.py 17:2 LF UTF-8 4 spaces Python 3.11 (PythonProject)

python_basics Version Control Current File Trial

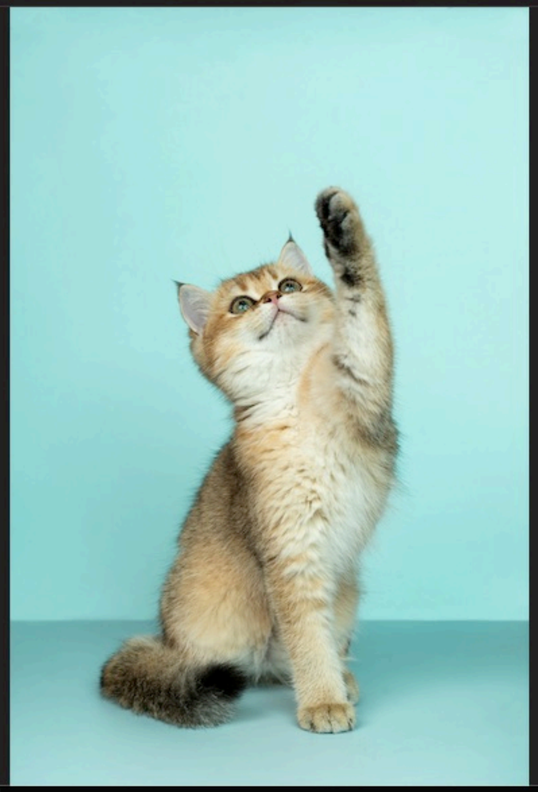
Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - enhance.py
  - explore_colors.py
  - img_rotated.png
  - intro.py
  - main.py
  - manipulate.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py | SoftwareTechnology1 | intro.py | manipulate.py | explore_colors.py | enhance.py

```python
# enhance.py
from PIL import Image, ImageEnhance

# import an image
image = Image.open('Assets/cat.jpg')

# create an enhancer
vibrance_enhancer = ImageEnhance.Color(image)
contrast_enhancer = ImageEnhance.Contrast(image)
brightness_enhancer = ImageEnhance.Brightness(image)
sharpness_enhancer = ImageEnhance.Sharpness(image)

# apply the enhancer
enhanced_image = sharpness_enhancer.enhance(1.5)

# show
image.show()
enhanced_image.show()
```

tmp_nwuyvly....



Run | enhance

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/enhance.py

Process finished with exit code 0
```

python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - enhance.py
  - explore_colors.py
  - image_filter.py
  - img_rotated.png
  - intro.py
  - main.py
  - manipulate.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py | SoftwareTechnology1 | intro.py | manipulate.py | explore_colors.py | enhance.py | image_filter.py

```python
#image_filter.py
from PIL import Image, ImageFilter

# create an image
image = Image.open('Assets/cat.jpg')

# apply a basic filter
#image_blur = image.filter(ImageFilter.BLUR)
#image_contour = image.filter(ImageFilter.CONTOUR)x
#image_emboss = image.filter(ImageFilter.EMBOSS)
#image_edge = image.filter(ImageFilter.FIND_EDGES)

# apply advanced filters
image_boxblur = image.filter(ImageFilter.BoxBlur(radius = 20))
image_gaussianblur = image.filter(ImageFilter.GaussianBlur(radius= 20))
image_unsharp = image.filter(ImageFilter.UnsharpMask(radius = 20))

image_boxblur.show()
image_gaussianblur.show()
```

Run image_filter

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/image_fi

Process finished with exit code 0
```

tmpcgzimb4n....

python_basics Version Control Current File Trial

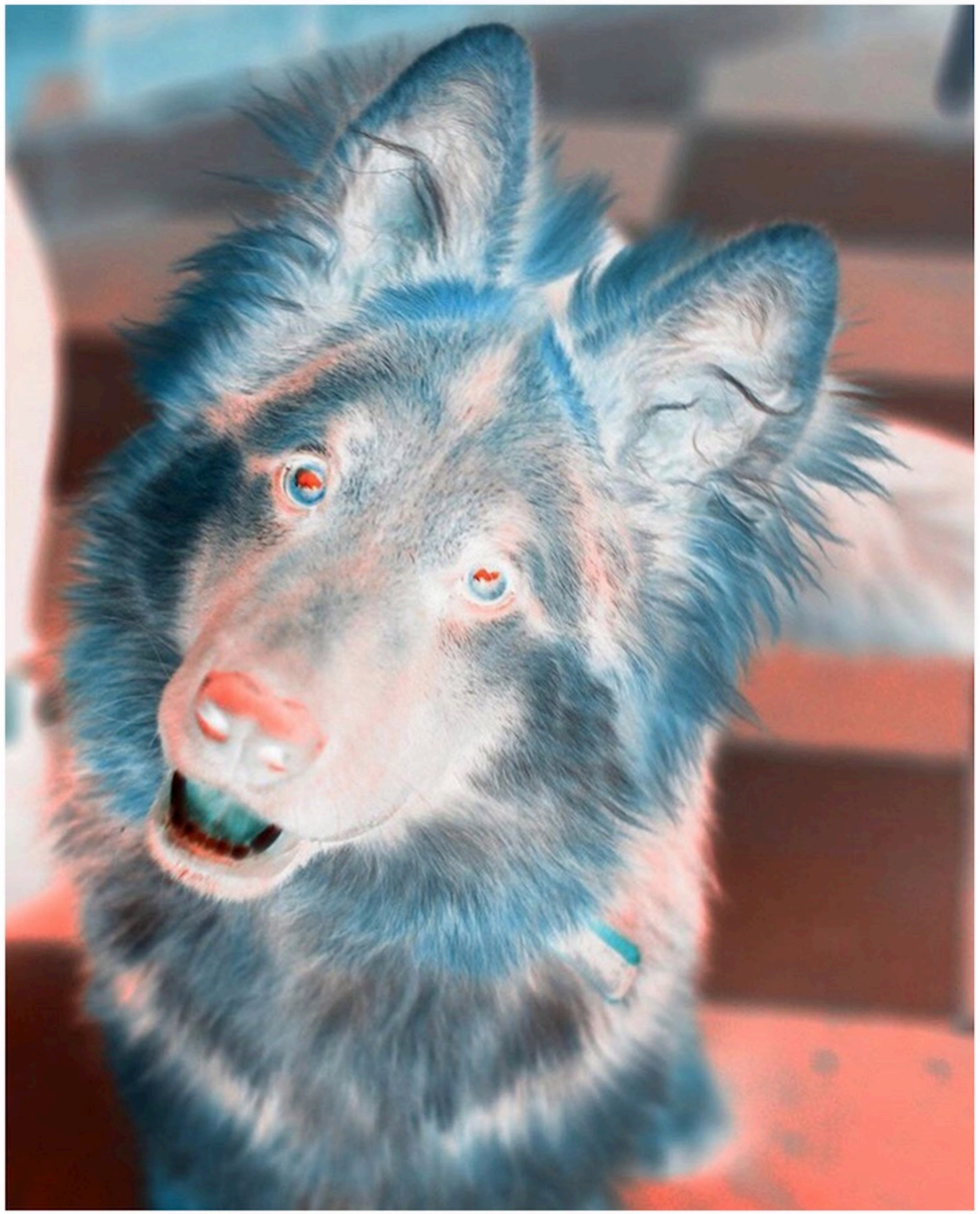
Project

- PythonProject [python_bas
  - .venv
  - Assets
  - Week_4_Lab_Activities
  - enhance.py
  - explore_colors.py
  - image_filter.py
  - image_operations.py
  - img_rotated.png
  - intro.py
  - main.py
  - manipulate.py
  - SoftwareTechnology1
- External Libraries
- Scratches and Consoles

main.py | SoftwareTechnology1 | intro.py | manipulate.py | explore_colors.py | enhance.py | image_filter.py | image_operations.py

```python
#image_operations.py
from PIL import Image, ImageOps

# create an image
image = Image.open('Assets/dog.jpg')

# position changes
image_mirror = ImageOps.mirror(image)
# image_scale = ImageOps.scale(image, 0.5)

# color changes
image_inverted = ImageOps.invert(image_mirror)

# add and remove
image_border = ImageOps.expand(
    image = image_inverted,
    border = 50,
    fill = (255,255,255))
# image_padded = ImageOps.pad(image, (4000,6000))
# image_crop = ImageOps.crop(image = image, border = 200)
image_border.show()
```

tmpck58tuuw....



Run image_operations

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/image_operations.py

Process finished with exit code 0
```

PythonProject > image_operations.py UTF-8 4 spaces Python 3.11 (PythonProject)